

Ahmed Abohadeed

AI Engineer

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Summary

Artificial Intelligence Engineer with a strong background in machine learning, deep learning, and generative AI technologies. Experienced in designing and deploying AI-driven applications using Python, LLMs, LangChain, RAG architectures, and modern backend frameworks. Adept at building intelligent systems, integrating AI models into enterprise workflows, and delivering scalable solutions that improve efficiency and user experience. Demonstrates excellent problem-solving abilities, technical adaptability, and a strong passion for advancing practical AI applications.

Education

Bachelor's degree from Faculty of Computers & Artificial Intelligence 07/2020 – 07/2024
Benha University | Graduation Project Grade: A+
• Graduation Project: Smart Driver Monitoring System (DMS) Monitored and Sponsored by Valeo
• Relevant Coursework: Artificial Intelligence, Machine Learning, Data Science, Python Programming, Data Structures & Algorithms, Embedded Systems, Database Management, Software Engineering

Professional Experience

Agentic AI internship, NTI 07/2026 – Present | Cairo , Egypt
• Built autonomous AI agents using LangChain and LangGraph.
• Developed RAG applications with vector databases and semantic search.
• Implemented tool calling and external API integrations for AI agents.
• Applied prompt engineering, reasoning, planning, and memory techniques to improve LLM performance.
• Designed secure AI systems with guardrails against prompt injection and jailbreak attacks.
• Deployed scalable AI applications following software engineering best practices.

AI & Software Engineer, Top Businesses 03/2026 – 06/2026 | Cairo , Egypt
• Designed and developed an AI-powered sales assistant using **Python**, **FastAPI**, and **Large Language Models (LLMs)** to automate customer interactions.
• Built an intelligent chatbot capable of understanding natural language queries and generating context-aware responses using **LangChain** and **Prompt Engineering**.
• Developed a **Retrieval-Augmented Generation (RAG)** pipeline to improve response accuracy by retrieving relevant business and product information.
• Integrated the AI assistant with **Odoo ERP** via **XML-RPC APIs** to automate product search, inventory lookup, customer management, and sales workflows.
• Designed and implemented scalable **RESTful APIs** to connect AI services with enterprise applications.
• Optimized AI workflows through intent classification, tool orchestration, and structured JSON outputs for reliable task execution.
• Developed multilingual conversational capabilities supporting both **Arabic** and **English** user interactions.
• Implemented vector-based semantic search to improve information retrieval and response relevance.
• Applied software engineering best practices including modular architecture, version control with **Git**, and API testing for maintainable, production-ready code.
• Collaborated with modern AI frameworks and open-source LLMs to evaluate model performance and improve system reliability.

Valeo (Mentorship Program), Valeo 07/2023 – 08/2024
• Developed production-grade Driver Monitoring System using computer vision and deep learning to enhance automotive safety and driver awareness
• Engineered custom CNN model for eye state detection achieving 98% validation accuracy and 97% training accuracy using PyTorch and TensorFlow
• Implemented real-time inference pipeline processing video frames at 30+ FPS with optimized MediaPipe integration for eye and head direction tracking
• Optimized system performance by converting MediaPipe to MediaPipe Lite, reducing model size by 60% while maintaining detection accuracy
• Ported Python codebase to C++ for embedded deployment on Raspberry Pi 4, ensuring low-latency performance in resource-constrained environment
• Integrated system with automotive HMI (Human-Machine Interface) for real-time data transmission, alerts, and driver feedback visualizatio

TECHNICAL SKILLS

Programming Languages: — Python, JavaScript, TypeScript, C++, C, SQL
Backend & Full Stack Development: — Flask, Django, FastAPI, RESTful APIs, React, Node.js, Express, PostgreSQL, MySQL, Redis,Docker ,Gradio,Streamlit
AI/ML & Data Science: — PyTorch, TensorFlow, Keras, OpenCV, MediaPipe, Scikit-learn, NumPy, Pandas, Matplotlib, NLP, Computer Vision, Model Fine-tuning(LORA &QLORA) , RAG , MCP ,MLOPS, Real-time Inference ,watsonxai
Databases: — : PostgreSQL, MySQL, SQLite, MongoDB, Database Design, Query Optimization, ORM (SQLAlchemy, Django ORM, Odoo ORM)
Odoo & ERP Development: — Odoo 17, Custom Module Development, Odoo ORM, XML Views, QWeb Templates, PostgreSQL Integration, Workflows & State Management, Security & Access Rights, Business Logic Implementation, CRUD Operations, Odoo Studio, ERP Customization, Debugging & Troubleshooting
Tools & Platforms: — Git, GitHub, GitLab, VS Code, Jupyter Notebook, Azure ML, Docker, Linux, Raspberry Pi, Postman

Selected Projects

Omerta.ai — NTI Graduation Project | AI-Powered Financial Crime Detection

- Developed an AI-powered platform for **fraud and financial crime detection**.
- Built **ML-based risk scoring** using transaction and behavioral signals.
- Integrated **LLMs, RAG,MCP, and Agentic AI** for fraud investigation and case analysis.
- Simulated realistic scenarios including **account takeover, suspicious transactions, VPN/proxy usage, and money transfers**.
- Designed a modular architecture using **Python, FastAPI, Machine Learning, and AI agents**.

AI-ERP Sales Agent

Built an AI-powered Telegram sales bot integrating Odoo 17 with multiple LLM providers (local Ollama, Groq, Google Gemini) to handle real-time customer conversations

- Designed a Retrieval-Augmented Generation (RAG) pipeline using sentence-transformers embeddings and ChromaDB for semantic product search over the catalog
- Architected a service layer (Odoo, Customer, Order services) with LRU caching on top of Odoo XML-RPC for real time inventory checks and quotation creation
- Implemented conversation memory management that persists and retrieves the last 5 messages per user to maintain context-aware dialogue
- Exposed a FastAPI backend with REST and Telegram webhook endpoints, containerized the system with Docker Compose alongside PostgreSQL and ChromaDB
- Developed an intent detection and JSON action parser to extract structured actions (e.g., create_order, get_stock) from LLM responses

Graduation Project: Driver Monitoring System

Leveraging computer vision and deep learning to detect eye states (open/closed) with 98% validation accuracy and 97% training accuracy. Implemented real-time inference in Python, utilizing Mediapipe for eye and head direction detection and optimizing it with Mediapipe Lite. Transferred the system to C++ for deployment on Raspberry Pi 4 and integrated it with an HMI (HumanMachine Interface) for data transmission and user interaction.

Multi-Modal RAG System

Developed an end-to-end multimodal RAG pipeline leveraging the **CLIP model** for cross-modal image-text embeddings and vector retrieval across text, PDF, and image data, enabling context-aware generation.

Video Captioning Pipeline

Engineered an automated video captioning pipeline using the **BLIP vision-language model** to generate descriptive captions from extracted frames and align them with video timestamps.

Certificates

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| • Build RAG Applications ,IBM | • Develop Generative AI Application , IBM | • Generative AI with Large Language Models ,DeepLearning.AI |
| • Microsoft Machine learning Engineer , DEPI | • Web Development ,IEEE | • Front_end using React JS, ITI |

Soft Skills

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| Problem Solving | Analytical Thinking |
| Adaptability | Continuous Learning |
| Time Management | Attention to Detail |
| Critical Thinking | Team Collaboration |